MATHEMATICS STANDARD
CBSE Examination Paper-2024

With Topper's Answer

Time allowed: 3 hours *Maximum Marks: 80*

General Instructions:

Read the following instructions very carefully and strictly follow them:

*(i) This question paper contains **38** questions. **All** questions are **compulsory.***

*(ii) This question paper is divided into **Five** Sections - **A, B, C, D** and **E**.*

*(iii) In **Section A** questions number **1** to **18** are Multiple Choice Questions (MCQs) and question number **19** and **20** are Assertion-Reason based questions of **1** mark each.*

*(iv) In **Section B** questions number **21** and **25** are Very Short Answer (VSA) type questions, carrying **2** marks each.*

*(v) In **Section C** questions number **26** to **31** are Short Answer (SA) type questions, carrying **3** marks each.*

*(vi) In **Section D** questions number **32** to **35** are Long Answer (LA) type questions, carrying **5** marks each.*

*(vii) In **Section E** questions number **36** to **38** are Case Study based questions carrying **4** marks each. Internal choice is provided in **2** marks questions in each case-study.*

*(viii) There is no overall choice. However, an internal choice has been provided in 2 questions in Section-**B**, 2 questions in Section-**C**, 2 questions in Section-**D** and 3 question in Section-**E**.*

(ix) Draw neat diagrams wherever required, Take $\pi = \frac{22}{7}$ wherever required, if not stated.

*(x) Use of calclator is **NOT** allowed.*

SECTION - A $20\times1=20$

This section consists of 20 questions of 1 mark each.

1. The value of k for which the system of equations $3x - y + 8 = 0$ and $6x - ky + 16 = 0$ has infinitely many solutions, is **1**

(A) –2 **(B)** 2 **(C)** $\frac{1}{2}$ **(D)** $-\frac{1}{2}$

Ans. (B) 2.

2. Point P divides the line segment joining the points A(4, –5) and B(1, 2) in the ratio 5:2. Co-ordinates of point P are **1**

(A) $\left(\frac{5}{2}, \frac{-3}{2}\right)$ **(B)** $\left(\frac{11}{7}, 0\right)$ **(C)** $\left(\frac{13}{7}, 0\right)$ **(D)** $\left(0, \frac{13}{7}\right)$

Ans. (C) $\left(\frac{13}{7}, 0\right)$

3. The common difference of an A.P. in which $a_{15} - a_{11} = 48$, is **1**

(A) 12 **(B)** 16 **(C)** –12 **(D)** –16

Ans. (A) 12

4. The quadratic equation $x^2 + x + 1 = 0$ has _____ roots. **1**

(A) real and equal **(B)** irrational **(C)** real and distinct **(D)** not-real

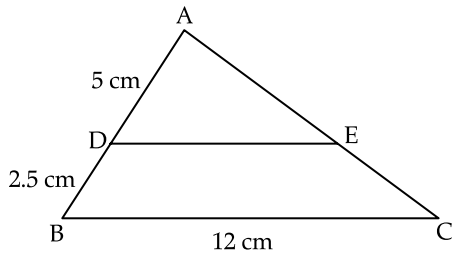
Ans. (D) not-real

5. If the HCF (2520, 6600) = 40 and LCM (2520, 6600) = 252 × k, then the value of k is **1**

(A) 1650 **(B)** 1600 **(C)** 165 **(D)** 1625

Ans. (A) 1650

6. In the given figure ΔABC is shown. DE is parallel to BC. If AD = 5 cm, DB = 2.5 cm and BC = 12 cm, then DE is equal to **1**



(A) 10 cm **(B)** 6 cm **(C)** 8 cm **(D)** 7.5 cm

Ans. (C) 8cm

7. If sin θ = cos θ, (0° < θ < 90°), then value of (sec θ . sin θ) is : **1**

(A) $\frac{1}{\sqrt{2}}$ **(B)** $\sqrt{2}$ **(C)** 1 **(D)** 0

Ans. (C) 1

8. Two dice are rolled together. The probability of getting the sum of the two numbers to be more than 10, is **1**

(A) $\frac{1}{9}$ **(B)** $\frac{1}{6}$ **(C)** $\frac{7}{12}$ **(D)** $\frac{1}{12}$

Ans. D) $\frac{1}{12}$.

9. If α and β are zeroes of the polynomial $5x^2 + 3x - 7$, the value of $\frac{1}{\alpha} + \frac{1}{\beta}$ is **1**

(A) $-\frac{3}{7}$ **(B)** $\frac{3}{5}$ **(C)** $\frac{3}{7}$ **(D)** $-\frac{5}{7}$

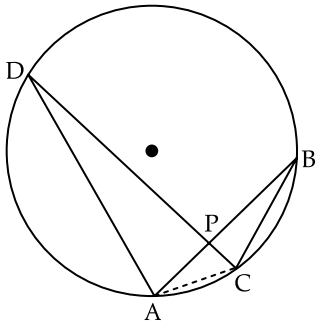
Ans. (C) $\frac{3}{7}$

10. The perimeters of two similar triangles ABC and PQR are 56 cm and 48 cm respectively. PQ/AB is equal to **1**

(A) $\frac{7}{8}$ **(B)** $\frac{6}{7}$ **(C)** $\frac{7}{6}$ **(D)** $\frac{8}{7}$

Ans. (B) $\frac{6}{7}$

11. AB and CD are two chords of a circle intersecting at P. Choose the correct statement from the following: **1**



(A) $\Delta ADP \sim \Delta CBA$ **(B)** $\Delta ADP \sim \Delta BPC$ **(C)** $\Delta ADP \sim \Delta BCP$ **(D)** $\Delta ADP \sim \Delta CBP$

Ans. (D) $\Delta ADP \sim \Delta CBP$

12. If value of each observation in a data is increased by 2, then median of the new data **1**

(A) increases by 2 **(B)** increases by 2n **(C)** remains same **(D)** decreases by 2

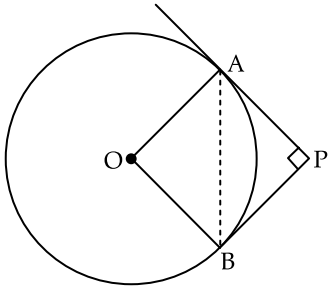
Ans. (A) increases by 2.

13. A box contains cards numbered 6 to 55. A card is drawn at random from the box. The probability that the drawn card has a number which is a perfect square, is **1**

(A) $\frac{7}{50}$ **(B)** $\frac{7}{55}$ **(C)** $\frac{1}{10}$ **(D)** $\frac{5}{49}$

Ans. (C) $\frac{1}{10}$

14. In the given figure, tangents PA and PB to the circle centred at O, from point P are perpendicular to each other. If PA = 5 cm, then length of AB is equal to **1**



(A) 5 cm **(B)** $5\sqrt{2}$ cm **(C)** $2\sqrt{5}$ cm **(D)** 10 cm

Ans. (B) $5\sqrt{2}$ cm

15. XOYZ is a rectangle with vertices X(–3, 0), O(0, 0), Y(0, 4) and Z(x, *y*). The length of its each diagonal is **1**

(A) 5 units **(B)** $\sqrt{5}$ units **(C)** $x^2 + y^2$ units **(D)** 4 units

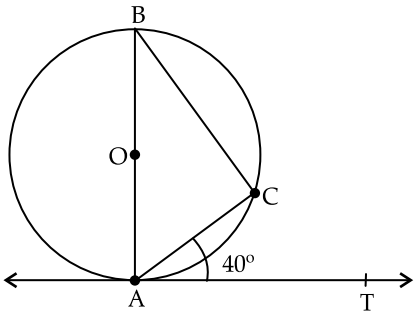
Ans. (A) 5 units

16. Which term of the A.P. –29, –26, –23, ..., 61 is 16 ? **1**

(A) 11th **(B)** 16th **(C)** 10th **(D)** 31st

Ans. (B) 16th

17. In the given figure, AT is tangent to a circle centred at O. If $\angle CAT = 40°$, then $\angle CBA$ is equal to **1**



(A) 70° **(B)** 50° **(C)** 65° **(D)** 40°

Ans. (D) 40°

18. After an examination, a teacher wants to know the marks obtained by maximum number of the students in her class. She requires to calculate _____ of marks. **1**

(A) median **(B)** mode **(C)** mean **(D)** range

Ans. (B) mode

Directions: In Question **19** and **20**, **Assertion (A)** and **Reason (R)** are given. Select the correct option from the following :

(A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

(B) Both Assertion (A) and Reason (R) are true. Reason (R) does not give correct explanation of (A).

(C) Assertion (A) is true but Reason (R) is not true.

(D) Assertion (A) is not true but Reason (R) is true.

19. **Assertion (A):** If $\sin A = \frac{1}{3}$ $(0° < A < 90°)$, then the value of $\cos A$ is $\frac{2\sqrt{2}}{3}$

Reason (R): For every angle θ, $\sin^2\theta + \cos^2\theta = 1$. **1**

Ans. (A) Both Assertion (A) and Reason (R) are true. Reason (R) is the correct explanation of Assertion (A).

20. **Assertion (A):** Two cubes each of edge length 10 cm are joined together. The total surface area of newly formed cuboid is 1200 cm^2.

Reason (R): Area of each surface of a cube of side 10 cm is 100 cm^2. **1**

Ans. (D) Assertion (A) is not true but Reason (R) is true.

SECTION - B

In this section, there are 5 questions of 2 marks each.

21. Can the number $(15)^n$, n being a natural number, end with the digit 0 ? Give reasons. **2**

Ans. If the number $(15)^n$, n being a natural number, were to end with the digit 0, then it would be divisible by 2 and 5.
This means it must have 2 among its prime factors.
However $(15)^n$ can only be factorised as $(3)^n \times (5)^n = (3\times5)^n$.
It does not have 2 among its factors.
The uniqueness of the 'Fundamental Theorem of Arithmetic' guarantees that the composition of 15 has only the prime nos 3 and 5.
Hence $(15)^n$ cannot end with the digit zero ever.

22. Find the type of triangle ABC formed whose vertices are A(1, 0), B(–5, 0) and C(–2, 5). **2**

Ans. Given : A(1, 0) ; B(-5, 0) ; C(-2, 5)
TO find : Type of triangle formed
Solution : Using the distance formula ie, $\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
we find the distance between each side.

$$AB = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$
$$= \sqrt{(1+5)^2+(0-0)^2}$$
$$= \sqrt{36}$$
$$= 6 \text{ units}$$

$BC = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
$= \sqrt{(-5+2)^2+(0-5)^2}$
$= \sqrt{(-3)^2+(-5)^2} \Rightarrow \sqrt{9+25}$
$= \sqrt{34}$ units

$CA = \sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$
$= \sqrt{(1+2)^2+(0-5)^2}$
$= \sqrt{(3)^2+(5)^2} \Rightarrow \sqrt{9+25}$
$= \sqrt{34}$ units

Since $\triangle ABC$ has 2 sides (CA and BC) of equal length (each $\sqrt{34}$ units)

we can conclude that $\triangle ABC$ is isosceles.

$\therefore \triangle ABC$ is an isosceles triangle.

23. (a) Evaluate : $2 \sin^2 30° \sec 60° + \tan^2 60°$. **2**

OR

(b) If $2 \sin (A + B) = \sqrt{3}$ and $\cos (A - B) = 1$, then find the measures of angles A and B. $0 \le A, B, (A + B) \le 90°$. **2**

Ans. (b) Given : $2(\sin A + \sin B) = \sqrt{3}$ and $\cos(A-B) = 1$

To find : values of A and B

Solution : Consider $2\sin(A+B) = \sqrt{3}$

$\Rightarrow \sin(A+B) = \sqrt{3}/2$

$\Rightarrow \sin(A+B) = \sin(60°)$

$A + B = 60° \longrightarrow (1)$

Also it is given, $\cos(A-B) = 1$

$\Rightarrow \cos(A-B) = \cos(0°)$

$\Rightarrow A - B = 0 \longrightarrow (2)$

now adding (1) and (2), $A + B + A - B = 60° + 0°$

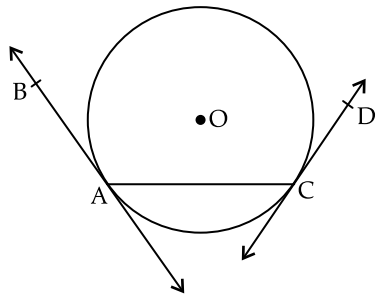
$2A = 60°$

$\therefore A = 30°$

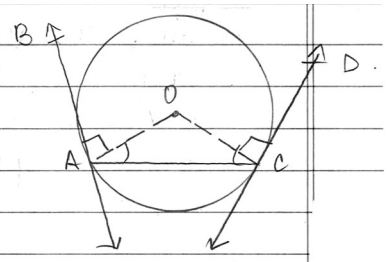
substituting value of A in (1), we get $B = 30°$

Thus values of A and B are 30° and 30° respectively.

24. In the given figure, AB and CD are tangents to a circle centred at O. Is $\angle BAC = \angle DCA$? Justify your answer. **2**



Ans.



Given: AB and CD are tangents to circle with centre O and chord AC.

To find: if $\angle BAC = \angle DCA$

Construction: Join OA and OC

Solution: Consider $\triangle OAC$ where

$OA = OC$ (radii of same circles)

so $\angle OAC = \angle OCA \rightarrow (1)$

(since angles opposite to equal sides in a triangle are equal)

We also know that a tangent to a circle is perpendicular to radius at the point of contact. Since AB and CD are both tangents,

$\therefore \angle OAB = \angle OCD \rightarrow (2)$ (as each is equal to 90°)

adding (1) and (2), $\angle OAC + \angle OAB = \angle OCA + \angle OCD$

$\angle BAC = \angle DCA$

hence proved that $\angle BAC = \angle DCA$.

25. (a) In what ratio is the line segment joining the points (3, –5) and (–1, 6) divided by the line $y = x$? **2**

Ans. (a) Let the points be A(3,–5) and B(–1, 6) divided by the point C(x, y) such that $x = y$.

section formula gives us the coordinates for points divided =

$$(m:n) = \left[\frac{(m)(x_2)+(n)(x_1)}{m+n}, \frac{(m)(y_2)(n)(y_1)}{m+n}\right]$$

$$\Rightarrow (K:1) = \left[\frac{(K)(-1)+(1)(3)}{K+1}, \frac{(K)(+6)+(1)(-5)}{K+1}\right]$$

$$\Rightarrow (x,y) = \left[\frac{-K+3}{K+1}, \frac{+6K-5}{K+1}\right]$$

thus x coordinate $= \left(\frac{-K+3}{K+1}\right)$ and y coordinate $= \left(\frac{+6K-5}{K+1}\right)$

but we know that $x = y$.

$$\frac{-K+3}{K+1} = \frac{+6K-5}{K+1}$$

$$3+5 = 6K+K$$

$$8 = 7K$$

$$K = \frac{8}{7}$$

hence the ratio is K:1 ie, 8:7

Required ratio is 8:7.

OR

(b) A(3, 0), B(6, 4) and C(–1, 3) are vertices of a triangle ABC. Find length of its median BE. **2**

SECTION - C

This section consists of 6 questions of 3 marks each.

26. (a) If the sum of first m terms of an A.P. is same as sum of its first n terms ($m \neq n$), then show that the sum of its first (m + n) terms is zero. **3**

Ans. (a) Given : $S_m = S_n$

To prove : $S_{(m+n)} = 0$

Proof : We know that sum of first 'x' terms in AP is given by –

$$\boxed{S_x = \frac{x}{2}[2a + (x-1)d]}$$

so $S_m = \frac{m}{2}[2a + (m-1)d]$

and $S_n = \frac{n}{2}[2a + (n-1)d]$.

but $S_m = S_n$ (given)

$\Rightarrow \frac{m}{2}[2a + (m-1)d] = \frac{n}{2}[2a + (n-1)d]$

$\Rightarrow m[2a + md - d] = n[2a + nd - d]$.

$\Rightarrow 2am + m^2d - dm = 2an + n^2d - dn$

$\Rightarrow 2am - 2an + m^2d - n^2d - dm + dn = 0$

$\Rightarrow 2a(m-n) + d(m^2 - n^2) - d(m-n) = 0$

$\Rightarrow (m-n)[2a + d(m+n) - d] = 0$

$\Rightarrow (m-n)[2a + d(m+n-1)] = 0$.

$\Rightarrow [2a + d(m+n-1)] = 0$ — (1)

now sum of first (m+n) terms will be,

$S_{(m+n)} = \frac{(m+n)}{2}[2a + (m+n-1)d]$.

but using (1) we get,

$S_{(m+n)} = \frac{(m+n)}{2}[0]$

thus $S_{(m+n)} = 0$

hence proved that sum of first (m+n) terms of this AP will definitely be zero.

OR

(b) In an A.P., the sum of three consecutive terms is 24 and the sum of their squares is 194. Find the numbers. **3**

27. Prove that $\sqrt{5}$ is an irrational number. **3**

Ans

Let us assume to the contrary that $\sqrt{5}$ is rational. Then it can be expressed in the form (a/b) where 'a' and 'b' are integers and co-primes. Also, $b \neq 0$.

So, $\sqrt{5} = \frac{a}{b}$

$\Rightarrow b\sqrt{5} = a$

On squaring both sides we get $(b\sqrt{5})^2 = (a)^2$

$\Rightarrow 5b^2 = a^2$

Since 5 divides a^2, then 5 divides a. (Since if 'p' a prime no. divides a^2, it surely divides a).

Now let $a = 5c$ (for any positive integer c).

Substituting, $5b^2 = a^2$

$\Rightarrow 5b^2 = (5c)^2$

$\Rightarrow 5b^2 = 25c^2$

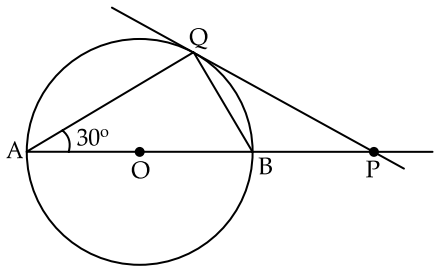
$\Rightarrow b^2 = 5c^2$

Here since 5 divides b^2, we know that 5 divides b.

However this means that 'a' and 'b' have a common factor 5, apart from 1. This contradicts the fact that they are co-primes. This contradiction arose due to incorrect assumption ie, that $\sqrt{5}$ is rational.

Hence, we conclude that $\sqrt{5}$ is irrational.

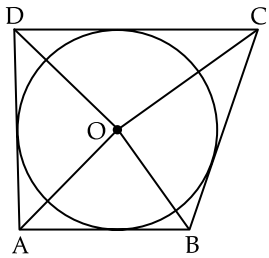
28. (a) In the given figure, PQ is tangent to a circle centred at O and $\angle BAQ = 30°$; show that BP = BQ. **3**



OR

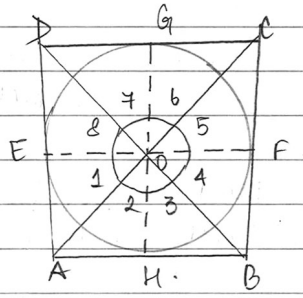
(b) In the given figure, AB, BC, CD and DA are tangents to the circle with centre O forming a quadrilateral ABCD. **3**

Show that $\angle AOB + \angle COD = 180°$



Ans.

16) Given : AB, BC, CD and DA are tangents to circle with centre O, forming quad. ABCD



To prove : $\angle AOB + \angle COD = 180°$

$\Rightarrow \angle 2 + \angle 3 + \angle 6 + \angle 7 = 180°$

Construction : Draw 2 diameters ie, EF and GH.
Join OG, OF, OH and OE.

Proof : We know that tangents drawn from an external point to a circle are equally inclined to the centre of the circle ie,

$\angle 1 = \angle 2$; $\angle 3 = \angle 4$; $\angle 5 = \angle 6$; $\angle 7 = \angle 8$ → (1)
(since AB, BC, CD and DA are tangents to same circle)

Also, sum of all angles around a point is 360°. So,

$\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 = 360°$

from (1), we get $2(\angle 2) + 2(\angle 3) + 2(\angle 6) + 2(\angle 7) = 360°$

$2[\angle 2 + \angle 3 + \angle 6 + \angle 7] = 360°$

$\angle 2 + \angle 3 + \angle 6 + \angle 7 = 180°$

$\angle AOB + \angle COD = 180°$

Hence proved that $\angle AOB + \angle COD = 180°$.

29. Prove that $\dfrac{1 + \sec\theta - \tan\theta}{1 + \sec\theta + \tan\theta} = \dfrac{1 - \sin\theta}{\cos\theta}$ **3**

Ans.

29. To prove : $\dfrac{1 + \sec\theta - \tan\theta}{1 + \sec\theta + \tan\theta} = \dfrac{1 - \sin\theta}{\cos\theta}$

Proof : Consider LHS.

$\Rightarrow \dfrac{(\sec\theta - \tan\theta + 1)}{(\tan\theta + \sec\theta + 1)}$

$\Rightarrow \dfrac{\sec\theta - \tan\theta + (\sec^2\theta - \tan^2\theta)}{(\tan\theta + \sec\theta + 1)}$ $\quad [\because 1 + \tan^2\theta = \sec^2\theta]$

$\Rightarrow \dfrac{(\sec\theta - \tan\theta)[1 + (\sec\theta + \tan\theta)]}{(\tan\theta + \sec\theta + 1)}$ $\quad [\because (a^2 - b^2) = (a-b)(a+b)]$

$\Rightarrow \dfrac{(\sec\theta - \tan\theta)(1 + \sec\theta + \tan\theta)}{(\tan\theta + \sec\theta + 1)}$

$\Rightarrow \sec\theta - \tan\theta$.

$\Rightarrow \dfrac{1}{\cos\theta} - \dfrac{\sin\theta}{\cos\theta} = \dfrac{1 - \sin\theta}{\cos\theta}$ ie, RHS.

thus LHS = RHS. Hence proved.

30. In a test, the marks obtained by 100 students (out of 50) are given below : **3**

Marks obtained :	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Number of students :	12	23	34	25	6

Find the mean marks of the students.

Ans.

Let assumed mean (A) for the data = 25

class interval (marks)	x_i [upper limit + lower limit / 2]	f_i (frequency)	d_i $(x_i - A)$	$f_i d_i$
0 – 10	5	12	$(5-25) \Rightarrow -20$	-240
10 – 20	15	23	$(15-25) \Rightarrow -10$	-230
20 – 30	A 25	34	$(25-25) \Rightarrow 0$	0
30 – 40	35	25	$(35-25) \Rightarrow 10$	250
40 – 50	45	6	$(45-25) \Rightarrow 20$	120
Total		100		(-100)

Now using the 'Assumed mean method', mean $= \left(A + \frac{\Sigma f_i d_i}{\Sigma f_i}\right)$

substituting the values, $25 + \left[\frac{-100}{100}\right]$

$\Rightarrow 25 - 1$

$\Rightarrow 24$

Thus the mean marks obtained by the students is 24.

31. In a 2-digit number, the digit at the unit's place is 5 less than the digit at the ten's place. The product of the digits is 36. Find the number. **3**

Ans.

Let the digit be in the form $(10x+y)$. So, digit in units place $= y$ and digit in tens place $= x$.

Now, we know $y = (x-5) \rightarrow$ ① [given]

also product of digits $= 36$.

$xy = 36$

$x(x-5) = 36$ [substituting value of y from ①]

$x^2 - 5x = 36$

$x^2 - 5x - 36 = 0$

by splitting the middle term we get,

$x^2 - 9x + 4x - 36 = 0$

$x(x-9) + 4(x-9) = 0$

$(x-9)(x+4)$

$\therefore x = 9$ or $x = -4$

We reject $(x = -4)$ as this number cannot be negative.

So, $x = 9$. from ①, we get, value of $y \Rightarrow (9-5) = 4$

Thus the number is $10x+y \Rightarrow 94$.

Hence the required number is 94.

SECTION - D

This section consists of 4 questions of 5 marks each.

32. (a) Using graphical method, solve the following system of equations : **5**

$3x + y + 4 = 0$ and $3x - y + 2 = 0$

Ans. (a) Consider the given equations.

(i) $3x + y + 4 = 0$

Required values of x and y are :

x	1	0	(−3)
y	(−7)	(−4)	5

(ii) $3x - y + 2 = 0$

Required values of x and y are :

x	1	0	(−2)
y	5	2	(−4)

[Lines are made on graph on page 23]

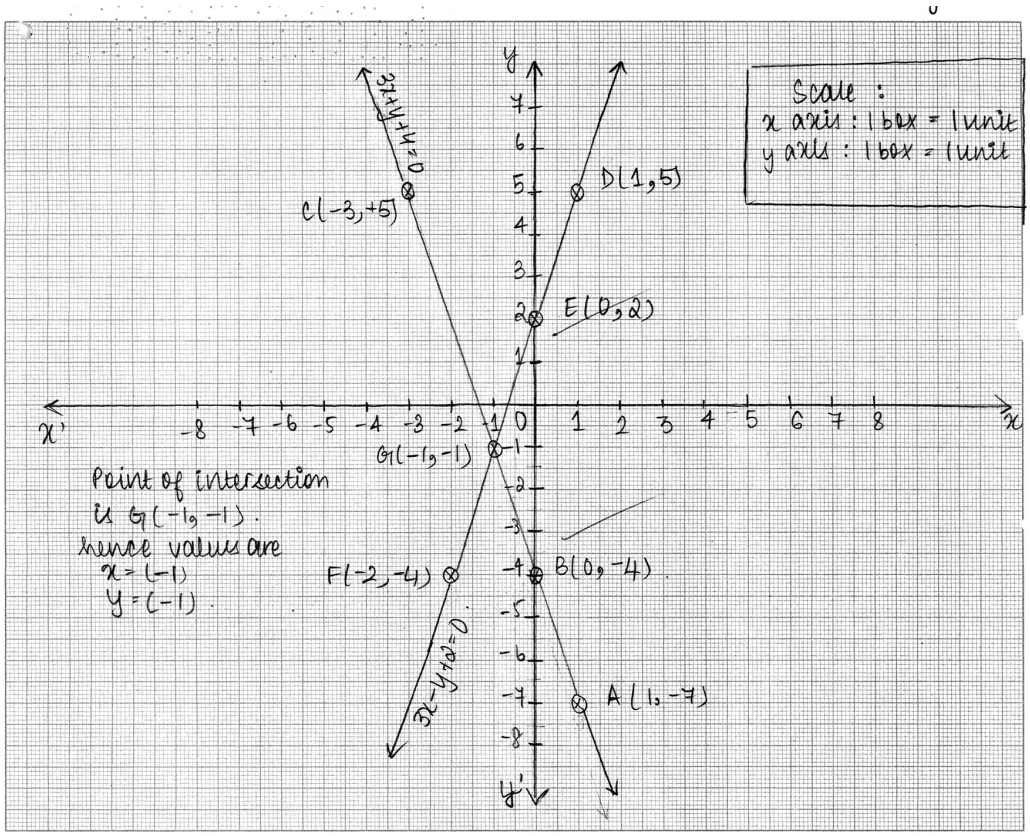
From page 23,

Point of intersection is G(−1, −1).

Hence value of $x = -1$ and $y = -1$.

We observe that this value satisfies each equation.

Hence solved.

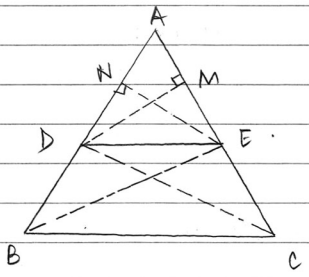


OR

(b) Tara scored 40 marks in a test, getting 3 marks for each right answer and losing 1 mark for each wrong answer. Had 4 marks been awarded for each correct answer and 2 marks been deducted for each wrong answer, then Tara would have scored 50 marks. Assuming that Tara attempted all questions, find the total number of questions in the test. **5**

33. (a) If a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then prove that the other two sides are divided in the same ratio. **5**

Ans. (a) **Statement:** If a line is drawn parallel to one side of a triangle to intersect the other 2 sides at two distinct points, the other two sides are divided in the same ratio.



Given: $\triangle ABC$ where $DE \parallel BC$ and DE intersects AB at D and intersects AC at E.

To prove: $\frac{AB}{BD} = \frac{AE}{EC}$

Construction: Draw $DM \perp AC$ and $EN \perp AB$. Join BE and AC.

Proof: Consider $\triangle ADE$ and $\triangle BDE$.

area $\triangle ADE$ = 1/2 x base x height $\Rightarrow$ 1/2 x AD x EN

area $\triangle BDE$ = 1/2 x base x height $\rightarrow$ 1/2 x BD x EN

$$\Rightarrow \frac{ar(\triangle ADE)}{ar(\triangle BDE)} = \frac{\frac{1}{2} \times AD \times EN}{\frac{1}{2} \times BD \times EN} = \boxed{\frac{AD}{BD}} \quad \text{— (1)}$$

Similarly, consider $\triangle ADE$ and $\triangle CDE$.

area $\triangle ADE$ = 1/2 x base x height $\Rightarrow$ 1/2 x AE x DM

area $\triangle CDE$ = 1/2 x base x height $\Rightarrow$ 1/2 x EC x DM

$$\rightarrow \frac{ar(\triangle ADE)}{ar(\triangle CDE)} = \frac{\frac{1}{2} \times AE \times DM}{\frac{1}{2} \times EC \times DM} = \boxed{\frac{AE}{EC}} \quad \text{— (2)}$$

Since $\triangle BDE$ and $\triangle CDE$ lie on the same base DE and between the same parallel lines (BC and DE), their area is equal. Hence (1) and (2) are equal.

from (1) and (2), $\boxed{\frac{AD}{DB} = \frac{AE}{EC}}$

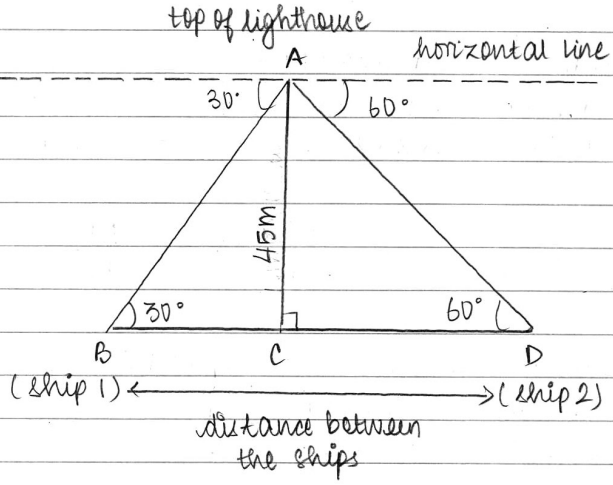
Hence proved.

OR

(b) Sides AB and AC and median AD to ΔABC are respectively proportional to sides PQ and PR and median PM of another triangle PQR. Show that ΔABC ~ ΔPQR. **5**

34. From the top of a 45 m high light house, the angles of depression of two ships, on the opposite side of it, are observed to be 30° and 60°. If the line joining the ships passes through the foot of the light house, find the distance between the ships. (Use $\sqrt{3} = 1.73$) **5**

Ans.



To find: Distance between ships ie, BD.

Solution: Consider △ABC where ∠B = 30° and ∠C = 90°

$$\tan A = \frac{AC}{BC}$$

$$\Rightarrow \tan 30° = \frac{45m}{BC}$$

$$\Rightarrow \frac{1}{\sqrt{3}} = \frac{45}{BC}$$

$$\therefore \text{Length of } BC = 45\sqrt{3}m \quad — (1)$$

4 1
1 7.3
× 6
1 0 3 . 8

now consider △ACD where ∠D = 60° and ∠C = 90°.

$$\tan 60° = \frac{AC}{CD}$$

$$\sqrt{3} = \frac{45}{CD}$$

$$CD = \frac{45}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{45\sqrt{3}}{3}$$

$$= 15\sqrt{3}m$$

so length of $CD = 15\sqrt{3}m$ — (2)

new length BD = BC + CD

$= 45\sqrt{3} + 15\sqrt{3}$ m (from ① and ②)

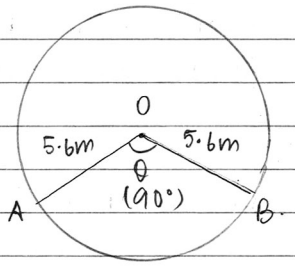
$= \sqrt{3}[45+15]$

$= (60)(1.73) = 103.8$ m

Thus distance between 2 ships is 103.8m or $\sqrt{3}(60)$m.

35. The perimeter of a certain sector of a circle of radius 5.6 m is 20.0 m. Find the area of the sector. **5**

Ans.



Given: circle with centre O and radius 5.6m.

Perimeter of sector OAB is 20m

To find: Area of sector OAB.

Solution: Consider sector OAB.

Perimeter of sector = 20m

2 radius + arc = 20m

$2(5.6) + \text{arc} = 20$

length of arc = 8.8m

length of arc $= \frac{\theta}{360^\circ} \times 2\pi r$.

$\Rightarrow \frac{\theta}{360^\circ} \times 2 \times \frac{22}{7} \times 5.6 = 8.8$

$\theta = \frac{8.8 \times 360 \times 7}{2 \times 22 \times 5.6}$

$= \frac{4 \times 18}{0.8} = 90^\circ$

Hence the required angle $(\theta) = 90^\circ$

Area of sector $= \frac{\theta}{360^\circ} \times \pi r^2$

Substituting values, $\frac{90}{4\ 360} \times \frac{22}{7} \times 5.6 \times 5.6\ 0.8$

$\Rightarrow \frac{22 \times 5.6 \times 8}{10\ 5}$

$\Rightarrow \frac{123.2}{5}$

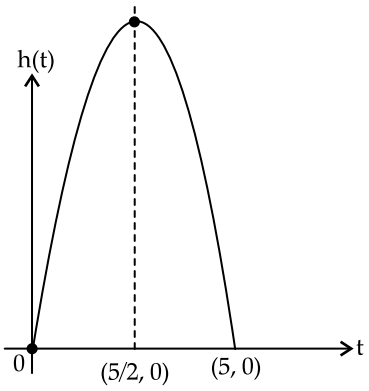
$\Rightarrow 24.64 m^2$

Thus the required area of the sector is $24.64 m^2$

SECTION - E

This section consists of 3 case based questions of 4 marks each.

36. A ball is thrown in the air so that t seconds after it is thrown, its height h metre above its starting point is given by the polynomial $h = 25t - 5t^2$.



Observe the graph of the polynomial and answer the following questions :

(i) Write zeroes of the given polynomial. **1**

(ii) Find the maximum height achieved by ball. **1**

(iii) (a) After throwing upward, how much time did the ball take to reach to the height of 30 m ? **2**

OR

(iii) (b) Find the two different values of t when the height of the ball was 20 m. **2**

Ans. 36. (i) Zeroes of the polynomial are given by the x-coordinate at the point of intersection with the coordinate axes.
Hence, zeroes of polynomial are 5 and 0.

(ii) The maximum height was achieved by the ball after 5/2 seconds.

$h = 25t - 5t^2$.

$= \left(25 \times \frac{5}{2}\right) - \left(5 \times \frac{25}{4}\right)$.

$\Rightarrow \frac{125}{2} - \frac{125}{4}$.

$\Rightarrow \frac{250 - 125}{4}$.

$\Rightarrow \frac{125}{4}$

$\Rightarrow 31.25\,m$

hence maximum height was 31.25m

(iii) (b) $h = 25t - 5t^2$

$20 = 25t - 5t^2$.

$0 = 25t - 5t^2 - 20$.

$5t^2 - 25t + 20 = 0$

$t^2 - 5t + 4 = 0$

$t^2 - t - 4t + 4 = 0$

$t(t-1) - 4(t-1) = 0$

$(t-1)(t-4)$

Thus the two different values of t are '1' and '4'.

37. The word 'circus' has the same root as 'circle'. In a closed circular area, various entertainment acts including human skill and animal training are presented before the crowd.

A circus tent is cylindrical upto a height of 8 m and conical above it. The diameter of the base is 28 m and total height of tent is 18.5 m.



Based on the above, answer the following questions :

(i) Find slant height of the conical part. **1**

(ii) Determine the floor area of the tent. **1**

(iii) (a) Find area of the cloth used for making tent. **2**

OR

(iii) (b) Find total volume of air inside an empty tent. **2**

Ans. 37. (i) $l = \sqrt{h^2 + r^2}$

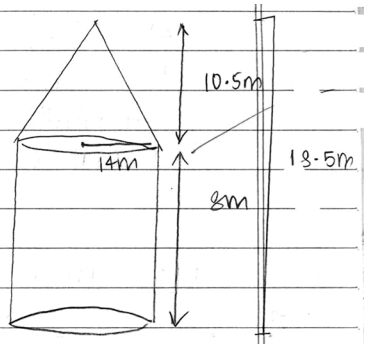
$= \sqrt{(10.5)^2 + (14)^2}$

$= \sqrt{110.25 + 196}$

$= \sqrt{306.25}$

$= 17.5\text{m}$

Hence the slant height is 17.5m



(ii) Floor area $= \pi r^2$

$= \pi(14)^2$

$= \frac{22}{7} \times 14 \times 14$

$= 44 \times 14$

$= 616\text{m}^2$

Thus floor area is 616m^2

(iii) (b) Volume = volume of cylinder + volume of cone

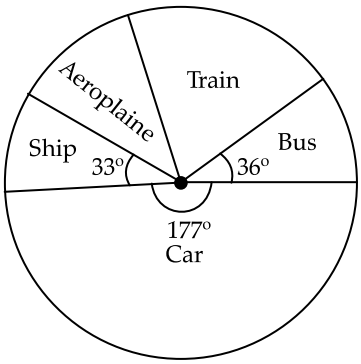
$= \pi r^2 h + 1/3 \pi r^2 h$

$= \pi r^2 (h + 1/3 h)$

$= \left[\frac{22}{7} \times 14 \times 14\right]\left[8 + \left(\frac{1}{3} \times 10.5\right)\right]$

$\Rightarrow 616(8+3.5)$
$\Rightarrow 616(11.5)$
$\Rightarrow 7084m^3$
Thus total volume is $7084m^3$

38. In a survey on holidays, 120 people were asked to state which type of transport they used on their last holiday. The following pie chart shows the results of the survey.



Observe the pie chart and answer the following questions :

(i) If one person is selected at random, find the probability that he/she travelled by bus or ship. **1**

(ii) Which is most favourite mode of transport and how many people used it ? **1**

(iii) (a) A person is selected at random. If the probability that he did not use train is $\frac{4}{5}$, find the number of people who used train. **2**

OR

(iii) (b) The probability that randomly selected person used aeroplane is $\frac{7}{60}$. Find the revenue collected by air company at the rate of ₹ 5,000 per person. **2**

Ans. 38. (i) $P(E)$ = Probability of bus or ship

$= \frac{\text{no. of outcomes favourable to E}}{\text{total possible outcomes of E}}$

$= \boxed{\frac{23}{120}}$

since no. of people who travel by bus or ship $= \frac{69}{360} \times 120 \Rightarrow 23$

Required probability is $\boxed{\frac{23}{120}}$

(ii) 'Car' is the favourite mode of transport.

No. of people who travel by it $= \frac{177}{360} \times 120$

$= 59$ people

Out of the 120 surveyed people, 59 use cars.

(iii) (b) No. of people who use planes = $\frac{7}{60} \times 120 \Rightarrow$ 14 people.

Thus, total revenue = ₹(5000)(14)

= ₹ 70,000

Thus total money made is ₹ 70,000

■■